

Work Log — 2026-08-03 to 08-09

LWFDSPC main controller | 10 commits, 35 files, +2092 / -169 | two internal releases tagged

Theme of the week: raising the speed-loop gains to fix uphill starting exposed several latent problems downstream — harsh deceleration, a creeping assist level 0, and a rollback counter that could never trigger. V0.18 and V0.19 address them.

Release	Date	Modbus ID03 0x15	Contents
v0.18	08-06	0x0018	EMB rollback net-edge counting, alarm code fix, LW_Jeff merge
v0.19	08-07	0x0019	e-lock, deceleration command filter, standstill speed fix

1 Bugs fixed

#	Problem	Cause and fix
1	Assist level 0 still crept forward at 1.04 km/h; reverse was not blocked at all	The level cap table's entry 0 is 0, but three <code>OUTPUT_MIN</code> floors downstream promoted it back to 1500 counts. Present in V0.16 too, masked by the EMB release threshold of 1600 ($1500 < 1600$, so the brake never released — the motor was instead stalling against a clamped brake). Lowering that threshold to 800 made the creep visible. Fixed by returning target 0 at level 0 before any floor applies, direction-agnostic so forward and reverse are blocked together, plus an EMB gate that prevents any command source from releasing the brake at level 0. Deliberately not an immediate hard lock, so dropping to level 0 while moving decelerates smoothly first. Now called e-lock .
2	Powered-rollback detection could never trigger uphill	The counter required <i>consecutive</i> reverse hall edges and reset to 0 on any same-direction edge. During a PI-versus-gravity tug of war the counter was zeroed every cycle while net displacement kept growing past the 1/4-wheel spec. Changed to net counting (reverse +1, forward -1, floor 0).
3	Rollback counter froze at standstill, causing a re-lock deadlock	The counter is only updated inside the hall-edge ISR, so with the vehicle clamped and still, nothing ran — not even the disarm path. It stayed at or above threshold, so re-applying the throttle re-locked on the next 20 ms tick before the wheel moved, and clearing the count requires motion. Now cleared on EMB LOCK and RELEASE.
4	Assist level index read past the array	Modbus delivers 4 bits (0–15) but the cap table has 6 entries and the index was never bounds-checked. Now clamped; levels 6–15 behave as level 5.
5	Standstill speed zeroing had never worked	<code>MotorStartSpeedPulses</code> was initialised to 0 and never assigned, so the guard was always false. X2CScope showed the effect: after a stop, a final hall edge decoded as reverse (brake-clamp rebound) held the measured speed at -900 counts for 840 ms until the timer-1 timeout. The command was already 0, so the speed loop saw a positive error and drove the motor forward — harmless only because UVW short and EMB usually clamped first. Set to 3 (0.19 km/h); the residue now clears within one 100 ms latch period.
6	A04 / A19 alarm codes swapped	EMB sensor fault was on A19 and motor hall fault on A04, the reverse of spec. Swapped, and the EMB fault bit added to the Modbus LCD error register.

2 New: deceleration command filter

The higher speed-loop gains faithfully amplified the command steps on throttle release and on level changes, which the rider felt as a jolt. The acceleration side already had rate-limit plus a first-order filter; the deceleration side went through the old step/time table. It now has its own filter, with **four independent quadrants**: forward/reverse × accel/decel.

Item	Detail
7 selectable decel curves	R from 20 to 100 counts/tick, i.e. 1200 ms down to 240 ms for full travel. Curves 4 and 6 match the old table's -25 and -40. Its selector is independent of the accel curve, since deceleration is safety-relevant and should not follow the rider's chosen accel level. Currently set to curve 2 (800 ms).
Rate limiter needed a down branch	It previously snapped straight to target whenever the target dropped, so a decel rate parameter would have had no effect whatsoever. This was the single most important implementation detail.
Min-RPM floor gated to accel only	On the decel path it would pin the command at 1500 counts, so the vehicle could never stop and UVW/EMB could never engage — the stop chain needs the command to be exactly 0.
Snap-to-zero threshold	The filter tail is asymptotic: about 240 ms at K=4 before the output rounds to 0, and at K ≥ 10 integer decay sticks at 1 count forever.
Reverse got its own parameters	Found that reverse had been silently using the <i>forward</i> accel curve — the filter call had no direction gate. Reverse 0→2.76 km/h was taking about 0.63 s against the 5–8 s its own step/time table intended, roughly 8× faster, with a preload jumping straight to 38% of reverse full speed. Defaults deliberately reproduce that existing behaviour, so reverse feel is unchanged this release and is now tunable.

The step/time tables and their ramp function are untouched and remain the fallback when the filter is disabled. Note they are still computed every cycle but no longer used on the live path, so tuning them now has no effect.

3 Tuning changes

Parameter	V0.16	V0.19	Reason
SPEEDCNTR_PTERM / ITERM	3000 / 10	5000 / 20	Uphill starting (from the LW_Jeff merge). Effective Kp +67%, Ki +100%
UVW_LOCK_STOP_PULSES	3	3	Went to 15 in V0.18, back to 3 in V0.19. At 15 the "stopped" threshold was 0.94 km/h, so EMB entered the clamp sequence while still rolling — the reported jolt and pad noise
EMBRAKER_RELEASE_THRESHOLD_FWD / REV	1600	800	Lets the brake release at lower commands; also what exposed bug #1
EMB_ROLLBACK_REV_EDGES	3	90	5.2 mm → 157 mm against the 159 mm quarter-wheel spec limit. With UVW back at 3 this is now the sole uphill rollback defence
EMB_ROLLBACK_CMD_THRESHOLD	500	100	Arming threshold for rollback detection, 0.35 → 0.069 km/h
LOGIC_TEMP_CONTROLLER_RATIO_NORMAL	0.33	0.35	Slightly wider normal current ceiling (levels 1–5 also raised ~0.02)
ACCEL_FILTER_CURVE_SELECT	4	5	Fastest accel curve, 2.76 km/h/s, 3.2 s to top speed

4 Documentation

Document	Purpose
版本及變動說明/	New folder; all 12 existing guides and change logs collected into it
V0.16_to_V0.19_version_variation	Full parameter-by-parameter comparison, bug list, test checklist, known open items
V0.18_customer_EMB_stop_and_accel_curve	Customer-facing: the four parameters for "EMB clamps too early" ranked by effect, and the 5 accel curves. No source paths
V0.18_change_summary	Internal change tables for V0.18
quick_tuning_sheet_V0.19	Re-versioned from V0.16

5 Verification status

Check	Result	Method
Full project build	clean	XC16 v2.10, project settings, zero diagnostics; .elf and .hex produced
Decel rate takes effect	confirmed	X2CScope: rate-limited ramp measured 390 ms against 375 ms predicted for that curve
Command reaches exactly 0	confirmed	X2CScope: <code>g_i16ScopeCmdOut</code> touches 0, snap step visible
On-vehicle testing of V0.19	not done	Braking distance, slope rollback, e-lock, level-0-while-moving all still pending

6 Open items

#	Item	Notes
1	V0.19 not road-tested	Highest priority: braking distance versus V0.18 (curve 2 is theoretically about 2.1× the old distance), slope rollback now that <code>EMB_ROLLBACK_REV_EDGES</code> = 90 is the only defence, and e-lock in both directions plus recovery back to level 3
2	EMB failsafe has no speed gate ✓ closed 08-13 — intentional, not a defect	<i>As observed this week:</i> on a downhill the UVW path cannot engage (wheel speed too high), Plan B depends on UVW, and rollback detection is disarmed once the command decays — so only the 3000 ms failsafe remains, and it clamps regardless of speed. Shortening it to 1000 ms was discussed but not applied at the time. Follow-up 08-13: the absence of a speed gate is deliberate design. If the command has been zero for the whole timeout and the vehicle is <i>still moving</i>, every normal deceleration and holding path has already failed — at that point continued motion is itself the hazard, so the vehicle must be stopped regardless of speed. The jolt and pad wear are an accepted cost. A speed gate must not be added for ride comfort, as it would disable the last line of defence exactly when it is needed. Comfort is instead addressed by letting an <i>earlier</i> path catch the case: the new downhill-slide detector (<code>EMB_DOWNHILL_*</code>, v0.20) clamps after 42 mm of travel, while kinetic energy is still negligible. <code>EMBRAKER_LOCK_TIMEOUT_MS</code> was separately shortened 3000 → 1000 ms on 08-10. See <code>s_logic_embraker.h</code> for the recorded intent
3	600-count steady-state gap at full speed	Command 12000 but measured speed plateaus at 11400 (7.87 versus 8.29 km/h). A PI with integral action should drive steady-state error to zero, so this is output saturation. Source not yet confirmed — bus voltage (25 V is known to give only about 7.1 km/h) or the temperature current clamp. Scope <code>piOutputOmega.out</code> against <code>outMax</code> to tell them apart
4	step/time tables are now dead on the live path	Still computed each cycle but unused. Tuning them has no effect — a trap for the next person
5	Dead parameters found	<code>LOGIC_THROTTLE_FAULT_DECEL_STEP_TIME</code> is never applied (the caller has no else branch); <code>UVW_LOCK_SPEED_THRESHOLD</code> and <code>UVW_LOCK_CURRENT_THRESHOLD</code> have zero references
6	Temperature current limit clamps a voltage	The variable is named as a current limit but is used to bound the <code>qVqRef</code> voltage reference. Possibly related to item 3

